Performance Analysis of UAVs Assisted Data Collection in Wireless Sensor Network

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***Abstract*—In the Internet of Things (IoT) services, the data of wireless sensor network needs to be collected. However, in the scenarios that have no infrastructure support, the data collection of sensors has great difficulty. Since unmanned aerial vehicle (UAV) has the characteristics of flexibility, it can be applied in the data collection for wireless sensor network (WSN). In this paper, we study UAVs supported data collection for WSN. Firstly, the entire region is divided into multiple cells. Secondly, the flight paths for single UAV and multiple UAVs are designed to cover all cells. The per-node capacity of sensor is derived, which is a function of the number of cells, the height of UAV, the number of sensors and the energy capacity of UAV. It is found that the per-node capacity with multiple UAVs is much larger than that with single UAVs. Then the optimal number of cells is derived to maximize the per-node capacity of WSN. Finally, we provide simulation results to verify our analysis. The discoveries in this paper may provide guideline for the UAVs assisted data collection in WSNs.**

***Index Terms*—Unmanned Aerial Vehicle; Wireless Sensor Network; Data Collection**

I. Introduction

Due to the high flexibility and maneuverability of UAVs, they are widely applied in the civil and military domains, such as environment monitoring, search and rescue, data collection, etc. [1]. As illustrated in Fig. 1, UAVs can act as mobile sink to collect the data of ground nodes in wireless sensor network (WSN). In [2], Cermakova *et al.* presented the entire process of data collection with UAVs, including data acquisition of entire geographic region, data preprocessing, classification and indices calculation [2]. Ma *et al.* in [3] studied the UAV supported data collection for mobile sensor network, and four fairness oriented algorithms of data collection are proposed.

The studies of UAVs acting as mobile sink focus on the path planning of UAV and scheduling of WSN. Rashed *et al.* in [4] studied the impact of flight paths on the UAV operation time and the coverage of WSN nodes. Ergezer *et al.* in [5] proposed an evolutionary path planning scheme to maximize the collected information of UAV. Zeng *et al.* in [6], presented a scheme to improve the energy efficiency of UAV communication by optimizing the flight path of UAV. In [7], Ho *et al.* proposed a heuristic path planing algorithm of UAV considering the energy consumption, delay of data collection and network lifetime. Besides, [7] proposed a cooperative relaying scheme for WSN nodes to improve the energy efficiency and network lifetime. In [8], Say *et al.* proposed priority-based frame selection for WSN nodes to improve the energy efficiency. Giorgetti in [9] proposed a cooperative transmission scheme of WSN to improve the network lifetime, communication reliability and throughout of UAV assisted data collection network.

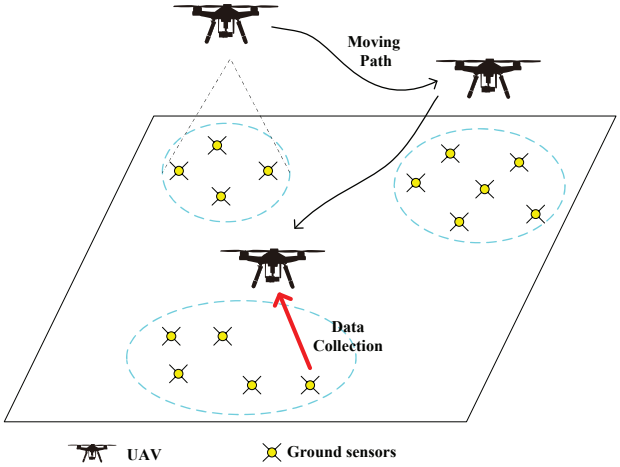

Fig. 1. UAV acts as mobile sink to collect data of WSN.

However, the previous literatures rarely studied the capacity of UAVs supported data collection network. Thus the impact of system parameters on the performance of data collection network is unknown. In this paper, we have derived the per-node capacity of WSN, which is a function of the number of WSN nodes, the size of the cluster of WSN nodes, the number of UAVs. The optimal size of the cluster of WSN nodes is achieved to maximize the per-node capacity. Moreover, we have discovered that multiple UAVs can achieve a much higher capacity than single UAV in the data collection network.

The remainder of this paper is organized as follows. Section II introduces the system model. In Section III, the single UAV supported data collection is studied. Section IV derives the per-node capacity of multiple UAVs supported data collection network. Section V provides the numerical results and analysis. Finally, we summarize this paper in Section VI.

II. System Model

The serving area of UAVs is assumed to be a square area with side length L, in which m sensor nodes are deployed. The area is divided into $n \times n$ small cells, which are squares with

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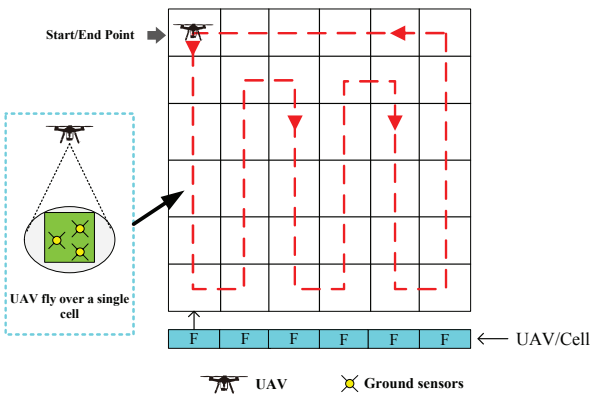

Fig. 2. The red line is the flight path of UAV, where each square represents a cell. When the UAV flies over the cell, the UAV will collect data from the sensor nodes within the cell.

side length $\frac{L}{n}$. In the single UAV scenario, a UAV flies along the path shown in Fig. 2, the similar flight path is proposed in [4], where UAV will periodically fly over a cell. When the UAV is over a cell, it will collect the data of the sensor nodes within the cell.

In the multiple UAVs scenario, the UAV flight path is as illustrated in Fig. 3, where multiple UAVs fly sequentially. When a UAV is above a ground cell, it will collect the data of the sensor nodes within the ground cell.

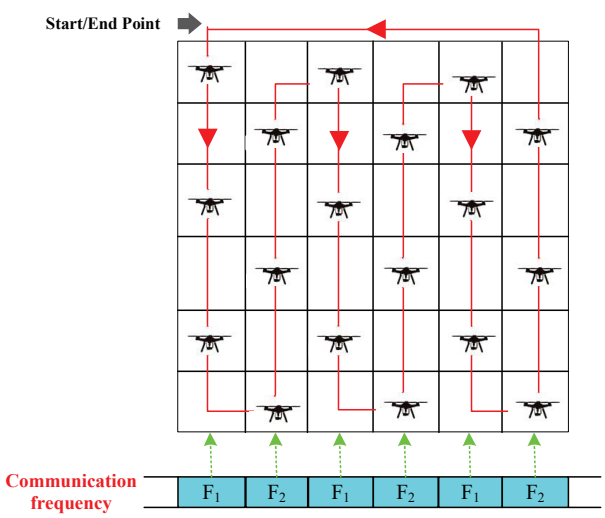

Fig. 3. The red line is the flight path of UAVs. The spectrum allocation scheme is shown in the bottom of the figure.

The per-node capacity λ denotes the amount of data that is accumulated at a sensor node and can be successfully collected by UAVs per unit time. Therefore the per-node capacity is related to the deployment of UAVs and the number of ground cells. When a UAV is above a cell, the capacity of the air-to-ground (A2G) channel is

$$C_{re} = W_u \log_2(1+\beta), \tag{1}$$

where C_{re} is the channel capacity, W_u is the channel bandwidth and β is the signal-to-noise ratio (SNR) as follows.

$$\beta = \frac{P_{d_0,d}{}^{-\alpha_d}}{N}, \tag{2}$$

where N is the power of additive white Gaussian noise (AWGN), $P_{d_0,d}$ is the power of signal received by UAV.

The A2G channel is modeled considering the occurrence probabilities of line-of-sight (LoS) and non-line-of-sight (N-LoS) components [10]. The path loss of NLoS propagation is higher than that of LoS propagation. Hence the received signal power of UAV is [11].

$$P_{d_0,d} = \begin{cases} P_d d_0^{-\alpha_d} & \text{LoS propagation,} \\ \eta P_d {d_0}^{-\alpha_d} & \text{NLoS propagation,} \end{cases} \tag{3}$$

where P_d is the transmit power of sensor node, d_0 is the distance between the UAV and the sensor node on the edge of the ground cell. Thus the worst A2G channel is considered. α_d is the path loss exponent of the A2G channel. η is additional attenuation factor due to NLoS propagation. Here, the probability of LoS propagationis as follows [11].

$$P_{Los} = \frac{1}{1 + A\exp[-B-(\theta - A)]}, \tag{4}$$

where A and B are constants that depend on the types of the environment, such as rural, urban, urban density, etc. θ is the elevation angle as follows.

$$\theta = \frac{180}{\pi} \times \sin^{-1}(\frac{h}{d_0}), \tag{5}$$

where $d_0 = \sqrt{h^2 + R_c^2}$, h is the flying altitude of the UAV and R_c is the coverage radius of the UAV. To cover the ground cell, R_c is set as $\frac{L}{\sqrt{2}n}$. Besides, we have $P_{NLos} = 1 - P_{Los}$.

III. Per-node Capacity in Single UAV Scenario

In this section, the per-node capacity of WSN is derived. Then the relation between the per-node capacity, the working time, energy constraint and the number of cells is analyzed.

A. Per-node capacity with single UAV

In the single UAV scenario, we assume that the UAV flies following the path in Fig. 2, where the UAV will periodically fly over a cell. Assuming that when the UAV flies over a cell, it will pause for time t_s to collect all the accumulated data of the sensors within the cell. Thus t_s is the amount of accumulated data of sensor nodes within the ground cell divided by the capacity of A2G channel C_{re}.

$$t_s = \frac{m\lambda(n^2 t_s + \frac{nL}{v})}{n^2 C_{re}} \Rightarrow t_s = \frac{mn\lambda L}{v(n^2 C_{re} - mn^2\lambda)}, \tag{6}$$

where v is the flying speed of UAV, n is an even number to realize the flight path in Fig. 2. In the single UAV scenario, the UAV can occupy the entire spectrum band, which is shown at the bottom of Fig. 2. Thus we rewrite (6) as follows.

$$t_s = \frac{m\lambda(n^2 t_s + \frac{nL}{v})}{n^2 W \log_2(1+\beta)} \Rightarrow t_s = \frac{m\lambda nL}{n^2 v[W\log_2(1+\beta) - m\lambda]}, \tag{7}$$

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$$\lambda = \frac{W(E_0 v - nLP_f)\log_2[1 + \frac{P_{Los}P_d\sqrt{h^2+\frac{L^2}{n^2}}^{-\alpha_d} + \eta(1-P_{Los})P_d\sqrt{h^2+\frac{L^2}{n^2}}^{-\alpha_d}}{N}]}{mE_0 v + mnLP_d}. \tag{9}$$

where W is the total spectrum bandwidth.

Since the energy of the UAV is limited, the UAV can not fly for a long time. When the energy of UAV is exhausted at the end point, the UAV can collect maximum data from ground sensors. The relation between the energy of the UAV E_0 and the pause time t_s is

$$E_0 = n^2 t_s P_d + \left(\frac{nL}{v} + n^2 t_s\right) P_f, \tag{8}$$

where P_f is the power consumption of UAV during flight. According to (3)(4)(5)(7), the relation between λ and n is derived in (9) on the top of this page.

If the impact of the angle θ on UAV is ignored, (2) can be simplified as follows.

$$\beta = \frac{P_d {d_0}^{-\alpha_d}}{N}. \tag{10}$$

Then (9) can be simplified as follows.

$$\lambda = \frac{W(E_0 v - nLP_f)\log_2[1 + \frac{P_d(\sqrt{h^2+\frac{L}{2n^2}})^{-\alpha_d}}{N}]}{mE_0 v + mnLP_d}. \tag{11}$$

B. Per-node capacity considering the randomness of nodes

In this section, m nodes are randomly distributed in n^2 cells $(m < n^2)$. UAV is not necessary to pause in every cell because there are empty cells. Thus the expectation of the number of empty cells needs to be derived. The probability that m sensor nodes are placed in k cells is

$$p(k) = \frac{S_2(m,k)A_{n^2}^k}{(n^2)^m}, \tag{12}$$

where $S_2(m,k)$ is the second type Stirling number, $A_{n^2}^k$ is the permutations number.

According to the definition of the second type of Stirling number [12], the recursive equation is

$$S(m+1,k) = S(m,k-1) + kS(m,k). \tag{13}$$

The closed form of the Stirling number is [12]

$$S_2(n,m) = \frac{1}{m!}\sum_{k=0}^{m}(-1)^k C_m^k (m-k)^n. \tag{14}$$

The expectation of the cells containing at least one sensor is as follows.

$$\overline{E} = \sum_{k=1}^{m} kp(k). \tag{15}$$

Thus the the expectation of the empty cells is

$$n_0 = n^2 - \overline{E}. \tag{16}$$

Similarly, when $m > n^2$, the expectation of the cells containing at least one sensor is

$$\overline{E} = \sum_{k=1}^{n^2} kp(k). \tag{17}$$

The (8) can be rewritten as

$$\lambda = \frac{n^2[W\log_2(1+\beta)](E_0 v - nLP_f)}{mnL(n^2-n_0)(P_d+P_f) + m(n^2-n_0)E_0 v} \tag{18}$$

C. Working time of single UAV

In the single UAV scenario, the UAV flies along the path shown in Fig. 2. The total length of the flight path is nL. Thus the flight time of UAV is $\frac{nL}{v}$. The total pause time is $n^2 t_s$. The working time T_w is

$$T_w = n^2 t_s + \frac{nL}{v}. \tag{19}$$

According to the energy constraints in (8), the working time of a UAV in (19) can be rewritten as follows.

$$T_w = \frac{vE_0 + nLP_d}{v(P_d + P_f)} \tag{20}$$

IV. Per-node Capacity in Multiple UAVs Scenario

In the previous section we discussed per-node capacity and working time in single UAV scenario. In this section we further studied the multiple UAVs scenario.

A. Per-node capacity with multiple UAVs

The per-node capacity of sensor node is a decreasing function of the interval time between two data collection processes. Thus, the scenario of multiple UAVs has been considered to reduce the interval time between two data collection processes and improve the per-node capacity. The UAV flight path is illustrated in Fig. 3. Multiple UAVs are sequentially dispatched and two adjacent UAVs are separated by a cell.

$$\lambda = \frac{2W[E_0 v - (n+1)LP_f]\log_2[1 + \frac{P_{Los}P_d\sqrt{h^2+\frac{L^2}{n^2}}^{-\alpha_d} + \eta(1-P_{Los})P_d\sqrt{h^2+\frac{L^2}{n^2}}^{-\alpha_d}}{N}]}{mE_0 v - mLP_f + mnLP_d} \tag{24}$$

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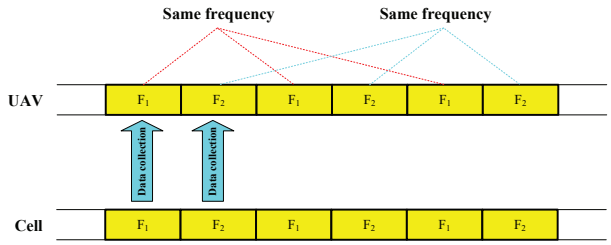

Fig. 4. Spectrum allocation in the scenario of Multiple UAVs

The UAVs will only serve the current cell with directional reception to avoid mutual interference. Besides, as illustrated in Fig. 4, the UAVs in the odd line use the band F_1 and the UAVs in the even line use the band F_2 to avoid the interference in the mutual coverage area of two UAVs.

According to (6), the pause time in the multiple UAVs scenario is

$$t_s = \frac{m\lambda(2t_s + \frac{2L}{nv})}{n^2 C_{re}} \Rightarrow ts = \frac{2m\lambda L}{v(n^3 C_{re} - 2nm\lambda)}. \tag{21}$$

In the spectrum allocation scheme of multiple UAVs scenario, each UAV can get a bandwidth of $W/2$. Thus we have

$$t_s = \frac{2m\lambda(2t_s + \frac{2L}{nv})}{n^2 W \log_2(1+\beta)} \Rightarrow \frac{4m\lambda L}{v[n^3 W \log_2(1+\beta) - 4nm\lambda]}. \tag{22}$$

Considering the new flight path for multiple UAVs scenario, The relation between the energy of the UAV E_0 and the pause time t_s is

$$E_0 = n^2 t_s P_d + (\frac{nL}{v} + \frac{L}{v} + n^2 t_s) P_f. \tag{23}$$

According to (3)(4)(5)(22)(23), the relation between the per-node capacity and the number of cells in the multiple UAVs scenario is derived as (24) shown in the bottom of page 3.

Similarly, if the impact of the angle θ is ignored, (24) can be simplified as follows according to (10).

$$\lambda = \frac{2W[E_0 v - (n+1)LP_f]\log_2[1 + \frac{P_d(\sqrt{h^2+\frac{L^2}{n^2}})^{-\alpha_d})}{N}]}{mE_0 v - mLP_f + mnLP_d} \tag{25}$$

B. Working time with multiple UAV

Compared to the scenario of single UAV, a returning path is added to avoid transmission collision of adjacent UAVs, which is illustrated in Fig. 3. So the total length of flight path of a UAV is $(n+1)L$. The working time T_w of a UAV is

$$T_w = n^2 t_s + \frac{nL}{v} + \frac{L}{v}. \tag{26}$$

According to the energy constraints in (23), the working time of a UAV is

$$T_w = \frac{E_0 v + nLP_d + LP_d}{v(P_d + P_f)}. \tag{27}$$

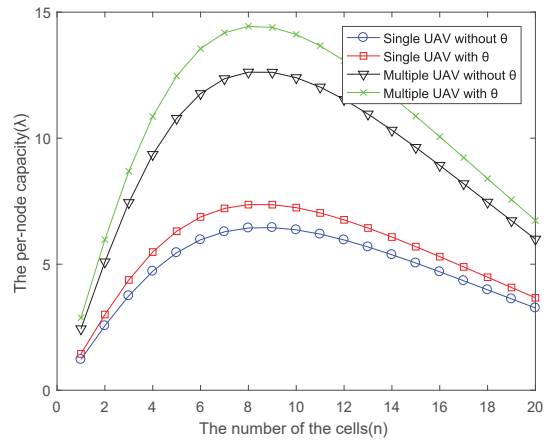

Fig. 5. The relation between the per-node capacity and the number of cells.

V. Simulation Results and Analysis

In this section, the typical results are verified by simulation results. Firstly, the relation between the per-node capacity and the number of cells in the single UAV and multiple UAVs scenarios is analyzed and compared. Then the relation between the working time and the number of cells is analyzed.

We assume that the UAV flight altitude is $h = 20\,\mathrm{m}$, the length of service area is $L = 400\,\mathrm{m}$, the number of sensor nodes $m = 100$, $E_0 = 1\,\mathrm{kJ}$, $P_f = 2\,\mathrm{w}$, the values of the constant A, B, α_d, $PLos$, η are 11.95, 0.136, 3, 0.8, 0.3, respectively. As illustrated in Fig. 5, the per-node capacity is a unimodal function of the number of cells. Thus there exists an optimal n that can maximize the per-node capacity λ. Besides, for the same number of cells, the per-node capacity with multiple UAVs is significantly higher than that with single UAV. These results can be explained. In practice, the service area should be divided into multiple cells to relieve the traffic load of UAV. However, when the number of cells is too large, the number of empty cells and the length of flight path will be large, which will decrease the per-node capacity of WSN. In the scenario of multiple UAVs, since the time interval between two adjacent UAVs is significantly decreased compared with the scenario of single UAV, the per-node capacity of WSN is improved.

When the random distribution of nodes is considered, the relation between the per-node capacity and the number of cells in the scenario of single UAV is provided in Fig. 6. It can be observed that when the UAV does not pause in the empty cell, the per-node capacity will be improved because the energy consumption along the flight path is reduced.

Fig. 7 illustrates the working time of the UAV acting as mobile sink. The working time of UAV increases with the increase of the number of cells. This is due to the fact that the increase of the number of cells will increase the length of flight path of UAV. Although the pause time of each UAV will decrease, its reduction can not make up for the increase of the flight time. Hence the total working hours of UAV will increase with the increase of the number of cells. The working time of UAV is directly determined by the energy of UAV.

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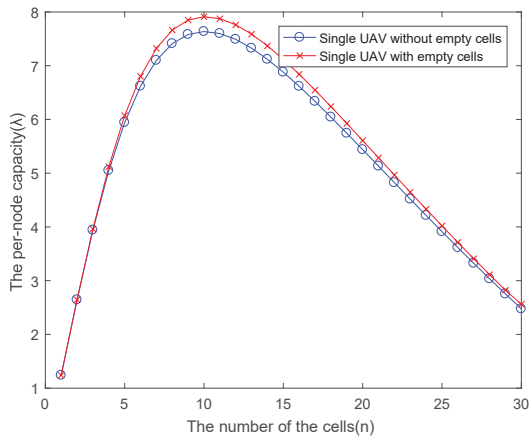

Fig. 6. The relation between the per-node capacity and the number of cells considering the empty cells in the scenario of single UAV.

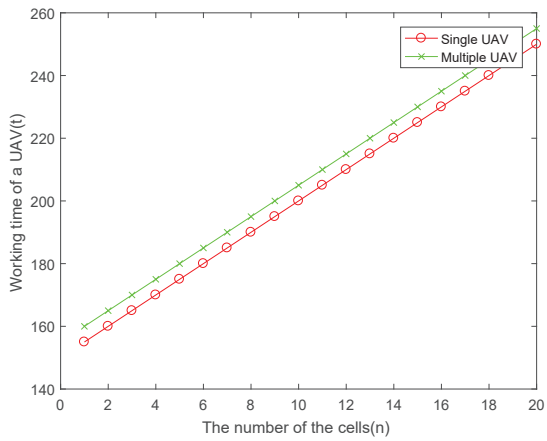

Fig. 7. The working time of a UAV in different scenarios.

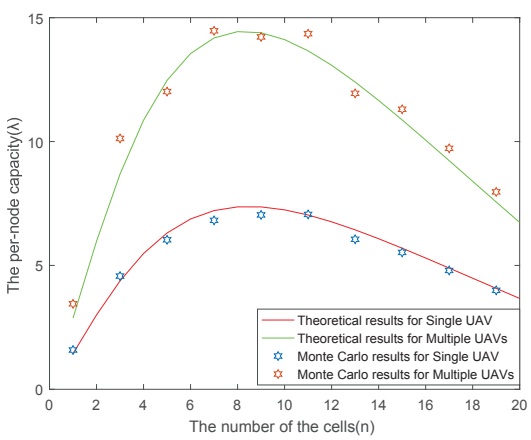

Fig. 8. The comparison between theoretical results and Monte Carlo results.

However, due to the extra returning path in the scenario of multiple UAVs, the working time of UAV in the multiple UAVs scenario is slightly larger than that in the single UAV scenario. Finally, the theoretical results and Monte Carlo results for the relation between the per-node capacity and the number of cells are provided in Fig. 8. Notice that the Monte Carlo results and the theoretical results match well, which verifies the theoretical analysis in this paper.

VI. Conclusion

This paper studied the UAVs assisted data collection in WSN. The per-node capacity of WSN and the working time of UAV are derived for the scenarios of single UAV and multiple UAVs. The theoretical results reveal that there exists an optimal number of cells to maximize the per-node capacity of WSNs. Besides, compared to the scenario of single UAV, the per-node capacity of WSN improves significantly with multiple UAVs. The simulation results have verified our theoretical analysis. Our work have proved that UAVs can enhance the performance of data collection in WSNs.

Acknowledgements

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